

Stationary Time Series

Definitions · Autocovariance · ACF & PACF · Wold

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Learning objectives

By the end of this chapter, you will be able to:

- state the definitions of *strict* and *weak (covariance) stationarity*;
- compute and interpret the *autocovariance* $\gamma(h)$ and *autocorrelation* $\rho(h)$ functions;
- read the *ACF* and *PACF* of an empirical series and identify candidate AR/MA orders;
- state the *Wold decomposition* and its consequences;
- run a *white-noise test* (Box–Pierce, Ljung–Box) to assess residuals.

1. Strict vs weak stationarity
2. Autocovariance and autocorrelation
3. Partial autocorrelation (PACF)
4. ACF / PACF signatures of standard processes
5. Wold decomposition
6. White-noise tests

Stationarity

1. Why stationarity?

Most classical time-series tools (estimation, forecasting, inference) require the data-generating process to have *stable statistical properties* over time. Without stationarity:

- the sample mean and variance lose their interpretation;
- ACF estimates become inconsistent;
- forecasts diverge and confidence intervals are wrong.

The decomposition step (Chapter 2) extracts trend and seasonality; the residual is what we hope is stationary.

1. Strict (strong) stationarity

Definition – Strict stationarity

A process $\{X_t\}_{t \in \mathbb{Z}}$ is *strictly stationary* if for every integer k , every $t_1 < \dots < t_k$ and every shift h , the joint distribution

$$(X_{t_1}, X_{t_2}, \dots, X_{t_k}) \stackrel{d}{=} (X_{t_1+h}, X_{t_2+h}, \dots, X_{t_k+h}).$$

- Strong: *all* finite-dimensional distributions are time-invariant.
- Implies stationarity of all moments (when they exist).
- Strict but rare; in practice we work with the weaker version.

1. Weak (covariance) stationarity

Definition – Weak stationarity

A process $\{X_t\}$ with $\mathbb{E}[X_t^2] < \infty$ is *weakly stationary* if there exist $\mu \in \mathbb{R}$ and $\gamma : \mathbb{Z} \rightarrow \mathbb{R}$ such that

$$\mathbb{E}[X_t] = \mu, \quad \text{Cov}(X_t, X_{t+h}) = \gamma(h) \quad \text{for all } t, h \in \mathbb{Z}.$$

- Constant mean, constant variance ($\gamma(0)$), and autocovariance depending only on the lag h .
- Strict $\Rightarrow$ weak (under finite second moments); the converse fails in general (it holds for Gaussian processes).

1. Strict vs weak stationarity

Property	Strict	Weak
Joint distribution invariant	yes	not required
Mean constant	yes	yes
Variance constant	yes	yes
$\gamma(h)$ depends only on h	yes	yes
Higher moments invariant	yes	not required
Useful for AR / MA / ARMA	yes (overkill)	yes

Throughout this course, “stationary” means *weakly stationary* unless otherwise specified.

1. White noise: the canonical stationary process

Definition – White noise

(ε_t) is a *white noise* of variance σ^2 (denoted $\text{WN}(0, \sigma^2)$) if:

$$\mathbb{E}[\varepsilon_t] = 0, \quad \text{Var}[\varepsilon_t] = \sigma^2, \quad \text{Cov}(\varepsilon_t, \varepsilon_{t+h}) = 0 \text{ for all } h \neq 0.$$

Stronger: *strong* white noise if the ε_t are i.i.d.

- ACF: $\rho(0) = 1$, $\rho(h) = 0$ for $h \neq 0$.
- No structure to exploit. Used as input to AR/MA models.
- In Python: `np.random.normal(0, sigma, n)` (Gaussian WN).

1. Random walk: a canonical *non*-stationary process

Definition – Random walk

$X_t = X_{t-1} + \varepsilon_t$ with $\varepsilon_t \sim \text{WN}(0, \sigma^2)$ and $X_0 = 0$. Equivalently $X_t = \sum_{s=1}^t \varepsilon_s$.

Why not stationary?

$$\text{Var}[X_t] = t \sigma^2 \rightarrow \infty, \quad \text{Cov}(X_t, X_{t+h}) = t \sigma^2 \quad (\text{depends on } t).$$

Variance grows linearly with t . Many financial price series behave like a random walk; Chapter 4 introduces tools to test for it (ADF, KPSS).

Autocovariance and ACF

2. Autocovariance and autocorrelation

Definition – ACVF and ACF

For a weakly stationary process with mean μ :

$$\gamma(h) = \text{Cov}(X_t, X_{t+h}) = \mathbb{E}[(X_t - \mu)(X_{t+h} - \mu)],$$

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)} \in [-1, 1].$$

Properties.

- $\gamma(0) = \text{Var}[X_t] \geq 0$.
- $|\gamma(h)| \leq \gamma(0)$ (Cauchy–Schwarz).
- $\gamma(-h) = \gamma(h)$ — even function.
- γ is a *positive-definite* function.

2. Sample ACF

Given observations $X_1, \dots, X_T$:

$$\hat{\gamma}(h) = \frac{1}{T} \sum_{t=1}^{T-h} (X_t - \bar{X})(X_{t+h} - \bar{X}), \quad \hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}.$$

Theorem – Bartlett's formula, white noise

Under $\text{WN}(0, \sigma^2)$, for fixed $h \geq 1$:

$$\sqrt{T} \hat{\rho}(h) \xrightarrow{d} \mathcal{N}(0, 1).$$

Practical bands. On the ACF plot, draw $\pm 1.96/\sqrt{T}$ bands; spikes outside are “significant” at the 5% level.

Partial autocorrelation (PACF)

3. Partial autocorrelation

Definition – PACF

$\alpha(h)$ is the correlation between X_t and X_{t+h} *after removing* the linear effect of $X_{t+1}, \dots, X_{t+h-1}$:

$$\alpha(h) = \text{Corr}(X_t - \hat{X}_t, X_{t+h} - \hat{X}_{t+h}),$$

where $\hat{X}_t$ and $\hat{X}_{t+h}$ are the linear projections on $\{X_{t+1}, \dots, X_{t+h-1}\}$.

- ACF: *total* correlation (direct + indirect).
- PACF: *direct* correlation only.
- Computed recursively (*Durbin–Levinson* algorithm).

ACF / PACF signatures

4. Theoretical ACF and PACF

Process	ACF $\rho(h)$	PACF $\alpha(h)$
White noise	$\rho(h) = 0$ for $h \neq 0$	$\alpha(h) = 0$ for $h \geq 1$
AR(p)	decays exponentially / sinusoidally	cuts off at $h = p$
MA(q)	cuts off at $h = q$	decays exponentially
ARMA(p, q)	decays after $h > q$	decays after $h > p$

Box-Jenkins identification. Plot the sample ACF and PACF; use the cutoff/decay patterns above to choose tentative (p, q) .

4. AR(1) example: $X_t = \phi X_{t-1} + \varepsilon_t$, $|\phi| < 1$

Theoretical autocorrelation:

$$\rho(h) = \phi^{|h|}.$$

- $\phi = 0.8$: $\rho(1) = 0.8$, $\rho(2) = 0.64$, $\rho(5) = 0.328$, $\rho(10) = 0.107$.
- $\phi = -0.8$: alternating sign, same magnitudes.
- PACF cuts off at lag 1: $\alpha(1) = \phi$, $\alpha(h) = 0$ for $h \geq 2$.

Stationarity of AR(1) requires $|\phi| < 1$. If $\phi = 1$, we obtain the random walk (non-stationary).

4. MA(1) example: $X_t = \varepsilon_t + \theta\varepsilon_{t-1}$

Theoretical autocorrelation:

$$\rho(0) = 1, \quad \rho(1) = \frac{\theta}{1 + \theta^2}, \quad \rho(h) = 0 \text{ for } h \geq 2.$$

- ACF *cuts off* at lag 1: clean signature.
- PACF *decays exponentially* (no cutoff).
- For $\theta = 0.5$: $\rho(1) = 0.4$.

Wold decomposition

5. Wold decomposition (1938)

Theorem

Every *purely non-deterministic* weakly stationary process X_t admits a unique decomposition

$$X_t = \mu + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}, \quad \sum_{j=0}^{\infty} \psi_j^2 < \infty,$$

with $\psi_0 = 1$ and (ε_t) a white noise of variance σ^2 (the *innovations*).

- Existence and uniqueness: any stationary process is an *infinite moving average* of innovations.
- Foundation of ARMA modelling: ARMA processes are finite-dimensional approximations of the Wold representation.

White-noise tests

6. Box–Pierce and Ljung–Box tests

Test H_0 : the residuals are white noise (no autocorrelation up to lag m).

Definition – Test statistics

$$Q_{\text{BP}} = T \sum_{h=1}^m \hat{\rho}(h)^2, \quad Q_{\text{LB}} = T(T+2) \sum_{h=1}^m \frac{\hat{\rho}(h)^2}{T-h}.$$

Under H_0 , $Q_{\text{LB}} \overset{a}{\sim} \chi_{m-p-q}^2$ where $p+q$ is the number of estimated ARMA parameters (or simply m if no model has been fitted).

- Reject H_0 if $Q_{\text{LB}} > \chi_{m-p-q, 1-\alpha}^2$.
- Choose m a few (10–40); for monthly data, $m \in \{12, 24, 36\}$.

6. Ljung–Box in Python

```
from statsmodels.stats.diagnostic import acorr_ljungbox

# residuals after decomposition / ARMA fitting
out = acorr_ljungbox(residuals, lags=[12, 24, 36], return_df=True)
print(out)
#          lb_stat  lb_pvalue
# 12          ...        ...
# 24          ...        ...
# 36          ...        ...
```

Small p -value (< 0.05) $\Rightarrow$ residuals are *not* white noise: there is structure left to model.

Chapter 4. Non-stationary time series: random walk, unit-root tests (ADF, KPSS), differencing, ARIMA.

- How to formally test for stationarity.
- Transformations to achieve stationarity (differencing, log).
- Integrated processes.

Thank you.

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